Mathematical overview

Model and first-visit facts. Use a finite connected undirected map, whole printed links and strictly positive prices. Buying pays once per link; traveling pays for every traversal. Length on paper is not price. With every place required, a cheapest network is a tree, with $n-1$ links on n places: a loop could lose a link and cost less. The first visit's other useful fact is the exchange certificate: a tree is cheapest exactly when no unused link is cheaper than a link on the tree route between its ends.

Cheapest networks solve every lowest-dearest-route problem at once. A route's *dearest* link is its largest price, not its total. Every route along a cheapest all-place network has the smallest possible dearest price among routes with those endpoints. If a cheaper-dearest route existed, remove a dearest tree link. That separates the tree into two sides; the proposed route crosses the separation on a cheaper link, giving a cheaper tree. Conversely, a tree with this property for *every* pair is cheapest, by the exchange certificate. Merely minimizing the largest price anywhere in the network is not enough.

Opening links determines the price list and cost. Open *all* links priced at most k, not just a chosen network. The reachable groups only merge. Two places first share a group at exactly their lowest possible dearest price. Let c_k be the number of groups and $c_0 = n$. Every cheapest tree has exactly $n - c_k$ links priced at most k: its low-price pieces must join every group, by the preceding route fact, and each piece is a tree. Thus it has $c_{k-1} - c_k$ links of price k. All cheapest trees share one price list even when their links differ. For positive *integer* prices,

$$\text{cheapest cost} = \sum_{k\geq 1} k(c_{k-1} - c_k) = \sum_{k\geq 0}(c_k - 1).$$

The sums stop when there is one group. Include thresholds at which nothing merges. This is an explanation of children's records, not a buying recipe to give them first.

Trips and networks react differently to repricing. Adding a to every price adds $a(n-1)$ to every all-place tree, so it changes no cheapest network while prices stay positive. A trip of j links gains aj, so different-length trips can overtake. Doubling preserves both kinds of optimum. More generally, strictly increasing repricing preserves all cheapest all-place trees, including ties, because it preserves every exchange comparison. Squaring positive prices can nevertheless change cheapest trips.

Optional places change the problem. Circles must connect; squares need not. A cheapest purchase has no loop and no optional leaf: either could be trimmed for a saving. With two required places it is a cheapest trip; with all places required it is the original tree problem. In between, a purchase can resist every cheaper single change yet not be cheapest. With $r \geq 2$ required places, at most $r-2$ optional places can have three or more bought links; the leaf-count proof is on guide p. 7. The all-place route, price-list, cost-from-groups, repricing and converse claims do *not* extend to optional places.

Discovery and evidence. K–1 should notice when two places first join a group and that a square can save, tie or cost extra (dots to 4, totals to 12). Grades 2–3 should connect the lowest dearest price to the first common ring, and notice that adding to every price can switch a trip while keeping the network (printed totals to 18). Grades 4–5 should explain these patterns by cuts and equal link counts, then test their limits with squares. A pattern seen on one map is a *conjecture*; a complete catalog proves that case, while cut and leaf arguments explain general claims. Incomplete searching proves no impossibility. Adults may read and record; children choose the links.

Preparation and shared launch

Status. Unpiloted and physically unrehearsed. Print scale, tape clearance, pawn fit, counter handling, preparation time and the proposed pacing still need a real rehearsal. Digital checks do not establish classroom readiness.

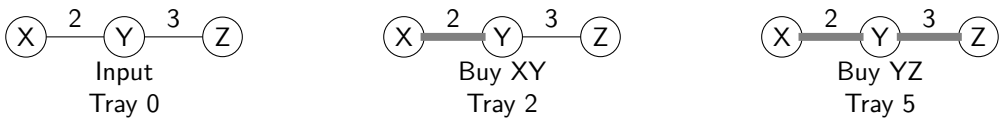
Kits and prints. Each pair or grades 4–5 trio reuses 30 counters (12–15 mm), ten removable tape strips (5×20 mm), a pawn or finger, a sleeve/pen for erasable rings, a tray and scrap paper for a second payment area. Two K–1 kits, two grades 2–3 kits, one grades 4–5 kit, plus 15 demonstration counters and four strips: **165 counters and 54 strips**. These are planning quantities, not attendance.

Print selected student pages once per pair/trio: `k-1.pdf` (7 pages), `grades-2-3.pdf` (6), `grades-4-5.pdf` (10), US Letter, single-sided, 100%. Cut nothing. Large maps are mats; small records/examples are pen-only. Working circles are 10 mm across, centers at least about 40 mm apart. No base page is needed; optionally print base `week-53-students.pdf` p. 3 (trap). There is no base materials PDF. Base guide: W53-guide-v4.

Most K–1 children are new: teach buying/paying. Many grades 2–3 children met only base pp. 1–4; do not assume swaps. Repeats: K–1 P1 is base P2; grades 4–5 P4 is base P4; grades 2–3 P5 and grades 4–5 P8/P9-bottom reuse base P6/P8's six-place map. Expect remembered answers; rings, price lists and trips are new questions.

Allow an unmeasured 15–20 minutes for prints/kits. Test strips on the six-place grid (grades 2–3 p. 4 or grades 4–5 p. 7), keeping prices visible and counters off roads; use a finger if a pawn hides a place. Write oversized trial totals, including the squared price 36. The kit covers optimal single payments, not every purchase: $18 + 15 = 33$ and $16 + 16 = 32$, so record one total and reuse its counters.

Whole group, 5 minutes including free handling. On the board use this non-task line (circles about 10 mm, centers at least 40 mm apart). Let a child mark one whole link and pay, then finish the line.



Say: "A network lets the pawn walk between every circled place on bought roads. On your maps, try to pay as few counters as you can. Trace X–Y–Z again without paying; then return a link and refund it. At each table demonstrate its next convention using the printed start/step/result: opening (K–1 p. 2; grades 2–3 p. 2; grades 4–5 p. 3), trips (grades 2–3 p. 3; grades 4–5 p. 6), or squares (K–1 p. 4; grades 2–3 p. 6; grades 4–5 p. 8). A trip uses all printed roads: write its word, then pay (XYZ costs 5; XYZY costs 8). Squares may be passed through but need not be visited. Introduce these separately.

Staffing and one hour. One adult per table; parent volunteer at K–1. Read K–1 P2 and grades 2–3 P3 pointing to pictures; K–1 children identify groups, adults may ring tiny records. Pool six-network catalogs; spare maps are workspace. Paraphrase grades 4–5 P10's "change" and Lee's P12 claim. The trio rotates chooser, payer/checker, recorder. Adults read/record, children choose.

Use 5 minutes movement, 5 handling/launch, 20 exploring, 5 break, 20 continuing, 5 sharing. **K–1: P1, P2, P4**, then P3 or P5; later visit P5–6, then P7–8 if ready. **Grades 2–3: P1–3**, then P4; later visit P5–8. **Grades 4–5: P1, P3, P7**, then P9 or P10; pool P2 or save it with P4. P5–6/P11–12 offer later explanations. Finishing every page is not the goal.

K–1 solutions and discretionary hints

AB means the whole link; buying it first or last does not make a new network. Spare record maps are workspace, not a promise of that many answers. Let children choose; the adult can read, count and draw their chosen links.

Problem 1 page 1

Minimum **4**, exactly {AB, AC, CD}, {AB, BC, CD}, {AC, BC, CD}. Reaching D costs at least 2, and two more links at least 1 each. Equality requires CD and two triangle sides; all three pairs work. *Hint if needed.* "Which road could reach D? Could the same payment buy a different triangle pair?"

Problem 2 page 2

With 1-dot links open, groups are **AB / C / D / EF**; up to 2 dots, **ABC / DEF**; up to 3 dots, **ABCDEF**; up to 4 dots, still one group. The pawn first reaches F from A at **3**. Include isolated places in rings. Opening means marking every eligible link; no payment total is requested. *Hint if needed.* "Where can the pawn go without crossing an unmarked road?" A group need not look round; its membership matters.

Problem 3 page 3

Minimum **9**: {AB, AC, CD, DE, EF} or {AB, BC, CD, DE, EF}. Price list $1, 1, 2, 2, 3$: AB/EF are forced cheapest exits; DE joins D/E; AC or BC joins C; CD is the cheapest crossing between ABC and DEF. Without 3- or 4-dot links those groups stay apart. *Hint if needed.* "Can you cross between the rings without a 3 or 4?"

Problems 4 and 5 pages 4 and 5

Each outer triangle has three links priced 4. Without S, any two cost **8**, the minimum. For P4, using S costs **6**, uniquely AS, BS, CS. For P5's top map, using S costs **9**, uniquely the three spokes: the square costs one extra. On P5's bottom map, using S costs **8**: AS, BS together with *one of* CS, AC or BC. The square ties rather than saves. Those are all three cheapest ways using S; adding the three outer pairs gives six cheapest overall.

Why complete: a cheapest join using S is a tree with three spokes, two spokes plus one outer link, or one spoke plus two outer links. Compare these forms, retaining the last even though it is wasteful when S is optional. *Hint if needed.* P4: "Must every circle use an outside road?" P5: "Must using the square mean buying all its roads?" Keep both payment piles.

Problem 6 page 6

Without X: **12**, any three outer sides (four choices). Using X: **8**, uniquely AX, BX, CX, DX. Three outer links cost 12; a five-place tree needs four links, each at least 2, and the four spokes attain 8. *Hint if needed.* "Could any other four-link purchase cost 8?"

K–1 continuation and grades 2–3 entry

K–1 Problem 7 page 7

Minimum **6**, uniquely AS, BS, ST, CT, DT. Neither square: 10 (AB, AC, CD); only S or only T: 9. With both, five links are needed, but only four price-1 links exist: $1 + 1 + 1 + 1 + 2 = 6$. *Hint if needed.* "How can the two small groups join through their squares?" A readiness-dependent continuation.

K–1 Problem 8 page 7

Use positive whole-number dots. A working answer is **2, 2, 4** on the three spokes; 1, 3, 4 and 2, 3, 3 also work, in any arrangement. Complete rule: sort the spoke prices $x \le y \le z$. The cheapest join using S costs $\min(x + y + z, x + y + 4, x + 8)$. It equals 8 exactly for $(1, 3, z)$ **or** $(2, 2, z)$ **with** $z \ge 4$**, or** $(2, 3, 3)$. Do not cap invented dots unless reducing the task deliberately. *Hint if needed.* "Could two square roads and one outside road beat your three roads?" A sum of 8 alone fails: 1, 1, 6 has a cost-6 alternative. The printed prompt leaves positivity implicit; say "at least one dot per road" before inventing prices.

Grades 2–3 solutions and discretionary hints

Problem 1 page 1

Minimum **9**. Every cheapest network uses CD and EF, any two of AB, AC, BC, and one of CE, DE. The complete six are:

AB, AC, CD, CE, EF	AB, AC, CD, DE, EF
AB, BC, CD, CE, EF	AB, BC, CD, DE, EF
AC, BC, CD, CE, EF	AC, BC, CD, DE, EF

A tree needs five links. The three usable price-1 links join ABC/EF; CD is D's cheapest exit; crossing ABCD/EF costs at least 4. Equality gives exactly the 3×2 listed choices. *Hint if needed.* "Which part is still separate? If you keep the right side the same, how many ways can you join A, B and C?"

Problems 2 and 3 pages 1 and 2

Pair	One lowest-dearest route	Dearest price	First common ring
A to D	ACD	2	2
B to D	BCD	2	2
D to F	DEF	4	4
A to F	ACEF	4	4

Other routes can be equally good; these problems do not request all routes. For P3: at price 1 the groups are **ABC / D / EF**; at 2, **ABCD / EF**; at 3, unchanged; at 4, one group. D is isolated below 2, and EF is separate below 4, proving the lower bounds. *Hint if needed.* P2: "Are we adding the prices or choosing the biggest?" P3: "Did the price-3 road actually join different groups?"

Grades 2–3 trips and optional places

Problem 4 page 3

On all three maps the cheapest network is AB, BC, CD. Their costs are **3, 6, 9** for original, $+1$ and $+2$ prices. Cheapest A–D trips are respectively **ABCD at 3; AD at 5; AD at 6**, each unique. These are the only two simple A–D routes; a repeated part costs extra. *Hint if needed.* "Which roads stayed the same even though their total changed?" Record links as well as costs, using the printed lines.

Problem 5 page 4

On all three maps the cheapest network is **AB, BC, CD, DE, EF**. Complete trip key:

Prices	Network cost	Every cheapest A–F trip	Trip cost
Original	8	ABCDEF, ACDEF	8 each
Every price 2 more	18	ACDF	15
Doubled	16	ABCDEF, ACDEF	16 each

For the network, AB, BC, CD, DE, EF are uniquely cheapest across cuts A / rest, AB / rest, ABC / DEF, ABCD / EF and ABCDE / F respectively. For trips, both original ways to reach C cost 3; then CDE costs 3 and EF costs 2. After $+2$, AC wins to C and DF wins from D to F. Guide p. 6 gives a complete shortest-trip certificate. *Hint if needed.* "Can a road we did not buy still be used on a trip?" Yes: a trip is on the whole printed map.

Problems 6 and 7 page 5

P6: write AB$= a$, BC$= b$, AC$= c$, using positive whole numbers. For a *strict* switch, $a + b < c < a + b + 2$, hence $c = a + b + 1$. For example 1, 1, 3; 1, 2, 4; 2, 2, 5. The two-link trip gains 4 while the direct trip gains 2. *Hint if needed.* "How much did each trip gain?" If a child allows tied cheapest trips, $a + b \le c \le a + b + 2$ also meets a weak reading of the prompt; ask whether they intended a tie or a clear switch, rather than rejecting it.

P7: **No; no**, for the no-square maps used so far and changes keeping prices positive. Every connecting tree has the same number of links, so adding the same amount preserves its ranking; doubling doubles every trip total. *Hint if needed.* "Do all networks gain the same number of extra counters? Do all trips?" P8 changes the rules, so the first answer need not survive there.

Problem 8 page 6

Left: **8 without S, 6 using S**. Right: **8 without, 9 using**. Without, choose any two outer links; using, the unique minimum is AS, BS, CS on each map. The three tree forms on guide p. 3 prove the comparisons. *Hint if needed.* "Is an optional place always a bargain?" Let the child demonstrate the two piles before discussing a general rule.

Grades 4–5: routes, groups and general claims

Problem 1 page 1

Use the complete grades 2–3 P2 key (guide p. 4), including its lower bounds. *Hint if needed.* "Which groups remain apart if that largest price is forbidden?"

Problem 2 page 2

Use grades 2–3 P1's complete six-network key and proof (guide p. 4), minimum **9**. *Hint if needed.* "Which links occur in every network? Where do they differ?" Pool cases; spare records here and in P4 are workspace.

Problem 3 page 3

Use grades 2–3 P3's groups and first-common thresholds (guide p. 4). Including the six isolated places before any link opens gives $c_0, \ldots, c_4 = \mathbf{6}, \mathbf{3}, \mathbf{2}, \mathbf{2}, \mathbf{1}$. *Hint if needed.* "If a road opens inside one ring, does the number of groups change?"

Problem 4 page 4

Minimum **6**, exactly {AB, AC, CD, DE}, {AB, BC, CD, DE}, {AC, BC, CD, DE}. Groups: at 1, **ABC / D / E**; at 2, **ABCDE**, unchanged thereafter. All optima have prices **1,1,2,2**: a four-link tree can use only two price-1 triangle sides, so costs at least 6; equality needs CD, DE and a triangle pair. *Hint if needed.* "Do three price-1 links help reach D or E?"

Problem 5 page 5

Sam is wrong. All four trees have largest price 3, but BC, CD, AD costs **9**; the other three cost **7** (AB and any two other sides). The unavoidable largest price does not determine total. *Hint if needed.* "Same dearest road, different payment piles?" Sam checks one number; P6 checks every pair.

Problem 6 page 5

Yes; no; yes, on a connected positive-price map with no squares. Every cheapest-network route is lowest-dearest; all cheapest networks have the same price list; threshold groups determine cost. The proof ideas open this guide. Two examples suggest, but do not prove, a universal statement.

For P2–3 the group drops are $6-3=3$, $3-2=1$, $2-2=0$, $2-1=1$. Therefore there are **three price-1 links, one price-2 and one price-4**, giving $3+2+4=9$. Alternatively add $5+2+1+1=9$. For P4, the drops are $5-3=2$, $3-1=2$, giving $2+4=6$, or $(5-1)+(3-1)=6$. Count the threshold-3 pause in the first map even though no link of that price is bought.

Hint if needed. "Suppose a cheaper-dearest route existed. What happens if we lift the dearest road in your tree?" Draw the two resulting sides and let the child find where the proposed route crosses. For the price-list question ask, "How many links are needed inside each ring?" Save the converse for P12.

Limit. P9's 2,2,4-spoke map has cheapest lists 4,4 and 2,2,4: optional squares change the theorem.

Grades 4–5 trips and optional places

Problem 7 page 6

Use grades 2–3 P4's complete network/trip key (guide p. 4). *Hint if needed.* "Which payment counts every bought road, and which counts only your trip?" Reuse the printed trip example before comparing totals.

Problem 8 page 7

Every cheapest network is **AB, BC, CD, DE, EF**, uniquely. The grades 2–3 P5 cut argument proves this. For the original, $+2$ and doubled rows, use its complete key (guide p. 4); the additional squared row is below.

Prices	Network cost	Every cheapest A–F trip	Trip cost
Each price times itself	14	ABCDEF	14

Complete trip certificate. The least costs from A to A,B,C,D,E,F are respectively

	A	B	C	D	E	F
original	0	1	3	5	6	8
$+2$	0	3	5	9	12	15
$\times 2$	0	2	6	10	12	16
squared	0	1	5	9	10	14

Every printed link satisfies $d(v) \le d(u) + w(uv)$ in both directions; summing along a route proves its cost is at least the destination label. The listed trips attain the bound. Equality requires each step to advance by exactly its price: originally C can follow either A or B, while D, E, F must follow C, D, E; after $+2$ the successful chain is A,C,D,F; doubling keeps the original choices; squaring forces A,B,C,D,E,F. Thus no cheapest trip is missing. These adult labels are not an extra child ledger.

None of the three changes alters cheapest *all-place* networks. Addition and squaring can change cheapest trips; doubling cannot. *Hint if needed.* "What does adding 2 do to a three-road trip compared with a five-road trip?" For squaring ask which costly single road now competes with several cheaper ones. A change that can alter the cheapest network is $p \mapsto 7 - p$: its minimum is 15, and AB, BC, CD, DE, EF is no longer cheapest. Price order matters.

Problem 9 page 8

Left star: cost 8, three ways, AB/AC, AB/BC, AC/BC; S is unused. A tree using S costs at least 9. **Right star: cost 8, six ways**: those same three outer pairs, plus {AS, BS, CS}, {AS, BS, AC}, {AS, BS, BC}. The three tree forms from guide p. 3 show completeness. **Bottom map: cost 8, two ways**, {AB, BC, CD, DE, EF} and {AC, CD, DE, EF}. Only A and F are circles; B,C,D,E are squares. A positive-price optimum for two required endpoints is just a cheapest simple trip, so P8's original trip list proves completeness. *Hint if needed.* "Which places must be reached here? Could you return a road that goes only to an optional dead end?"

Problem 10 page 9

Cheapest is **6**, uniquely AS, BS, CS. From AB and BC, the answer is **no**: with a cap of 8, a purchase with no spoke can only be an outer pair. Returning a link disconnects a circle; buying any link exceeds 8; swapping an outer link for one spoke still disconnects a circle. Only swaps to the other outer pairs survive. The same holds with cap 9. A cap of **10** is enough: buy AS (10), swap AB for BS (8), swap BC for CS (6). *Hint if needed.* "Could one new spoke connect both sides left by returning an outside road?" A swap is one atomic exchange; its final state must connect the circles. This optional-place example, not a contradiction, marks the limit of the first-visit exchange test.

Grades 4–5 continuations, explanations and sources

Problem 11 page 10

Minimum **6**, uniquely AS, BS, ST, CT, DT, as in K–1 P7 (guide p. 3). A square place cannot have just one bought link in an unrestricted cheapest purchase: that link is a positive-price dead end, removable without separating any required circle. A square can be unused (degree zero); do not confuse that with an optional leaf in the bought tree.

With four circles, at most **2** squares can have three or more bought links; S and T attain it. Generally ignore unused places. The bought tree has only required leaves. If there are L leaves and branch degrees $d_i \geq 3$, counting link ends gives

$$-L + \sum_i (d_i - 2) = -2, \qquad L = 2 + \sum_i (d_i - 2).$$

Degree-2 places contribute zero; each branch contributes at least one. Hence at most $L - 2 \leq r - 2$ branches when $r \geq 2$ places are required. With $r = 4$, at most two are optional branches. *Hint if needed.* "When a path splits, how many extra ends must appear?" Drawn branches can explain the bound; this map alone proves only attainability.

Problem 12 page 10

Lee is right, on a connected positive-price map with no squares. First test P5's cost-9 purchase BC, CD, DA: its A–D–C–B route has dearest price 3, but AB costs 1, so it fails Lee's test. Sam's one number for the whole network misses this contrast.

Each unused link f is itself an available route, so Lee's condition says f costs at least every link on the tree route between its ends. This is the base week's exchange certificate, which proves global cheapestness; see W53-guide-v4, p. 7 for the transformation proof. *Hint if needed.* "What does Lee's condition say about a route consisting of just one unused road?" Enact one exchange before returning to the general explanation.

Sources consulted and boundaries

Reviser checked the final student PDFs W53-RV-K/M/U-v1, outline, draft, reviews and base guide W53-guide-v4 (exchange proof p. 7). The writer also consulted student sources, base research notes, Week 8 return-visit and Week 98 model guides. Use final-PDF numbering, not old page-review numbering.

The outline's background, not newly consulted originals: lowest-dearest routes (grades 2–3 P2–3; grades 4–5 P1/P3/P6) from Hu, "The Maximum Capacity Route Problem," *Operations Research* 9(6), 1961, pp. 898–900 (outline author checked abstract only), and Wikipedia's "Widest path problem"; repricing (grades 2–3 P7; grades 4–5 P8) from Sedgewick/Wayne, *Algorithms*, 4th ed., §4.3 companion, Exercise 1; Sam (grades 4–5 P5) from its Web Exercise 1; squares (grades 4–5 P9–11) from "Steiner tree problem." Hwang/Richards/Winter, *The Steiner Tree Problem* (1992), is cited background, not consulted.

Base notes trace the activity to CS Unplugged Activity 9, "The Muddy City," printed pp. 76–80/PDF pp. 1–5, and *Math Circle by the Bay*, "How we teach," p. ix/PDF p. 10. Those originals were unavailable. Elementary proofs and finite checks are supplied here. Materials and pacing remain untested; record actual use and handling issues.